Leadcrystal

Energy Evolved.

Lead Crystal Energy Solutions

Next-generation lead crystal battery systems for a sustainable future. High-performance, high-safety, and 99% recyclable.

LEAD CRYSTAL® BATTERIES

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POWERED BY
Betta Batteries

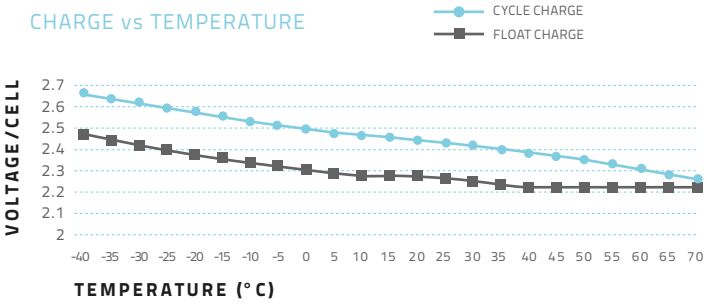
SPECIFICATION

Nominal Voltage		12V	
Rated Capacity (10 hour rate)		7.2 AH	
Dimension	Total Height (top of terminal)	100 mm	3.94 "
	Height	94 mm	3.7"
	Length	151 mm	5.94"
	Width	65 mm	2.56"
Weight		Approximately 2.3 kg / 5.07 lbs	
Capacity 25°C	120 hour rate (70mA)	8.4 AH	
	20 hour rate (390mA)	7.8 AH	
	10 hour rate (720mA)	7.2 AH	
Internal Resistance	Fully charged Battery (25°C)	30mΩ	
Self-Discharge 25°C	Capacity after 3 month storage	95%	
	Capacity after 6 month storage	85%	
	Capacity after 12 month storage	80%	
Max Discharge Current 25°C	72A (5S)		
Terminal	Standard	F1	
	Optional	F2	
Charging (Constant Voltage)	Cycle	Initial Charging Current 2.16A 14.7V/ (25°C)	
	Float	13.6V/ (25°C)	

DISCHARGE CURRENT AND END VOLTAGE

Discharge current (A)	End voltage (V)
0.05C or below or Intermittent discharge	11.4
0.05C of current close to it	11.1
0.1C of current close to it	10.8
0.2C of current close to it	10.5
From 0.2C to 0.5C	10.2
From 0.5C to 1C	9.6
From 1C to 3C	9.0
Current in excess of 3C	7.8

CHARGE vs TEMPERATURE



CHARGE vs TEMPERATURE CHART

temperature	-40	-35	-30	-25	-20	-15	-10	-5	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70
Cycle Charge	2.66	2.64	2.62	2.60	2.58	2.56	2.54	2.52	2.50	2.48	2.47	2.47	2.45	2.45	2.43	2.41	2.39	2.37	2.35	2.33	2.31	2.29	2.27
Float Charge (voltage/cell)	2.46	2.44	2.42	2.40	2.38	2.36	2.34	2.32	2.31	2.30	2.29	2.29	2.29	2.27	2.26	2.24	2.23	2.23	2.23	2.23	2.23	2.23	2.23

CONSTANT CURRENT DISCHARGE CHARACTERISTICS: UNITS AMPERES (25°C)

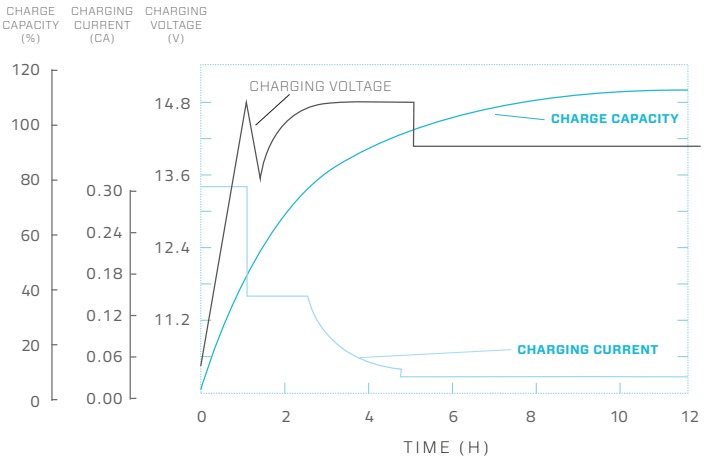
End Voltage per cell	5min	15min	30min	45min	1h	2h	3h	4h	5h	6h	8h	10h	12h	20h	24h
1.60V	26.42	13.98	8.45	6.16	4.96	2.84	2.06	1.62	1.38	1.17	0.89	0.74	0.63	0.41	0.33
1.67V	24.55	13.52	8.32	6.12	4.95	2.83	2.03	1.61	1.36	1.17	0.89	0.74	0.62	0.41	0.33
1.70V	24.30	13.32	8.24	6.03	4.91	2.81	2.01	1.60	1.34	1.15	0.89	0.74	0.62	0.40	0.33
1.75V	22.27	12.90	8.16	5.99	4.83	2.75	2.01	1.58	1.33	1.14	0.89	0.73	0.62	0.40	0.33
1.80V	19.98	12.07	7.82	5.83	4.70	2.71	2.00	1.58	1.31	1.13	0.88	0.72	0.62	0.39	0.33
1.83V	19.09	11.07	7.70	5.62	4.49	2.68	1.92	1.51	1.28	1.09	0.86	0.69	0.59	0.38	0.33
1.85V	17.89	10.74	7.20	5.41	4.37	2.58	1.87	1.49	1.25	1.05	0.85	0.68	0.58	0.38	0.32

DISCHARGE DATA WITH CONSTANT POWER UNITS: WATTS PER CELL (25°C)

End Voltage per cell	5min	15min	30min	45min	1h	2h	3h	4h	5h	6h	8h	10h	12h	20h	24h
1.60V	44.15	24.55	15.81	11.53	9.27	5.37	3.92	3.11	2.63	2.26	1.74	1.44	1.21	0.81	0.66
1.67V	42.03	24.18	15.17	11.44	9.28	5.37	3.87	3.10	2.63	2.25	1.74	1.44	1.21	0.81	0.66
1.70V	41.78	24.01	15.17	11.44	9.20	5.33	3.87	3.09	2.58	2.23	1.73	1.42	1.20	0.80	0.66
1.75V	38.91	23.72	15.18	11.44	9.15	5.28	3.86	3.09	2.58	2.22	1.72	1.41	1.20	0.80	0.66
1.80V	35.70	22.51	14.86	11.24	9.11	5.28	3.85	3.08	2.56	2.22	1.71	1.41	1.20	0.78	0.66
1.83V	34.46	20.68	14.73	10.90	8.74	5.24	3.75	2.98	2.53	2.15	1.71	1.36	1.18	0.77	0.65
1.85V	31.92	20.22	13.69	10.49	8.49	5.12	3.64	2.94	2.46	2.11	1.65	1.35	1.16	0.77	0.65

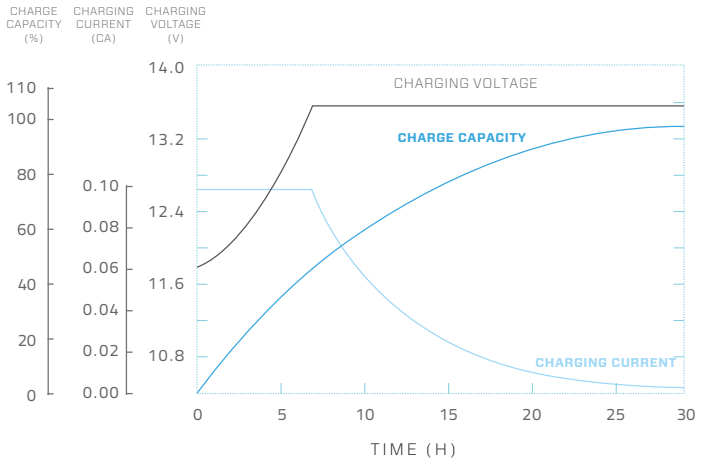
CYCLE CHARGE CHARACTERISTIC (25°C)

REGULAR CYCLE CHARGE CHARACTERISTICS 77°F (25°C)



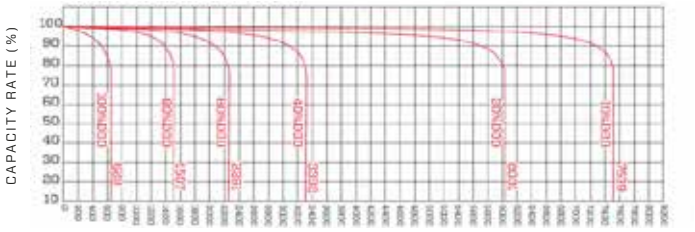
FLOATING CHARGE CHARACTERISTIC (25°C)

FLOATING CHARGE CHARACTERISTICS 77°F (25°C)

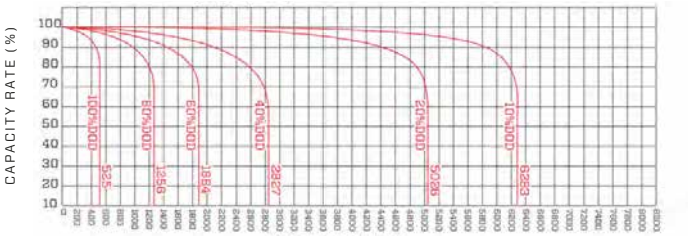


CYCLE LIFE CURVE GRAPH

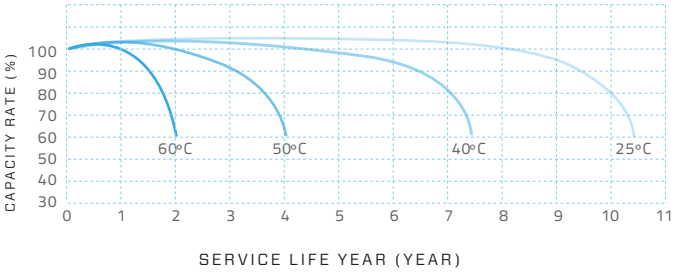
CYCLE LIFE CURVE GRAPH (25°C)



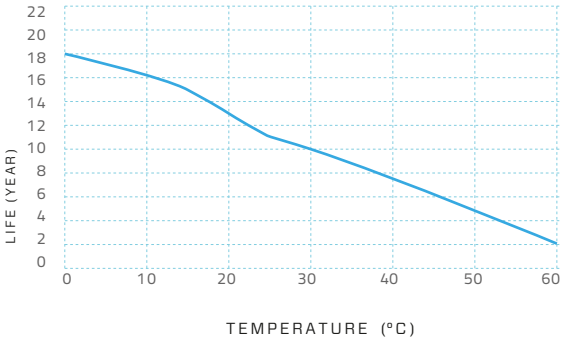
CYCLE LIFE CURVE GRAPH (40°C)



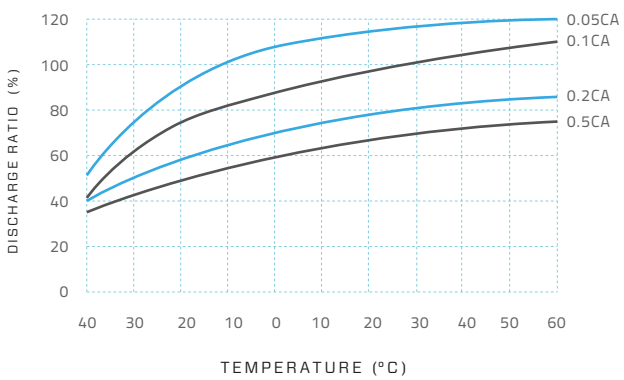
TEMPERATURE & FLOAT SERVICE LIFE



FLOAT SERVICE LIFE CURVE GRAPH



TEMPERATURE & DISCHARGE CAPACITY



6-CNFJ-7.2 12V / 7.2 A

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Performance: Robust, resilient, high performing. Lead Crystal® batteries can be discharged deeper, cycled more often (also in extreme temperatures) and have a longer service life. They recover to full rated capacity over and over again.

Technology: A unique micro-porous high absorbent mat (AGM), high-purity lead calcium selenium plates, safe Si02 electrolyte solution that solidifies into a white crystalline powder when charged/discharged.

Cleaner & Safer: Less acid, no cadmium, no antimony. Lead Crystal® batteries are up to 99% recyclable and are classified as non-hazardous goods for transport.

Markets: Lead Crystal® batteries are being used in telecoms, UPS, petrochem/marine, defence, renewable energy, health care, manufacturing, transportation and electric motion (wheelchairs, golf carts & trolleys).

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